

Time Series Fundamentals

This notebook explains the basic concepts of Time Series in a simple way.

We will cover:

- What is Time Series
- Types of Data
- Patterns in Time Series
- Components of Time Series
- Time Series Models

What is Time Series?

Time Series is data collected over time in a sequence.

- The order of data matters
- Each value depends on time

Examples:

- Daily temperature
- Stock prices
- Sensor readings

What is Time Series Data?

Time Series Data is a dataset where observations are recorded at regular time intervals.

Examples of intervals:

- Hourly
- Daily
- Monthly
- Yearly

Key Point:

Time is the most important feature in this type of data.

Cross-Sectional Data



Data collected at a single point in time across multiple entities.

- No time dependency

Examples:

- Income of 100 people in 2024
- Marks of students in one exam

Pooled (Panel) Data

Combination of Time Series and Cross-Sectional Data.

- Same entities tracked over time

Examples:

- Salary of employees over multiple years
- Sales of multiple stores over months

Types of Data Comparison

Data Type	Description
Cross-Sectional	Multiple entities at one time point
Time Series	One entity over multiple time points
Pooled (Panel)	Multiple entities over multiple time

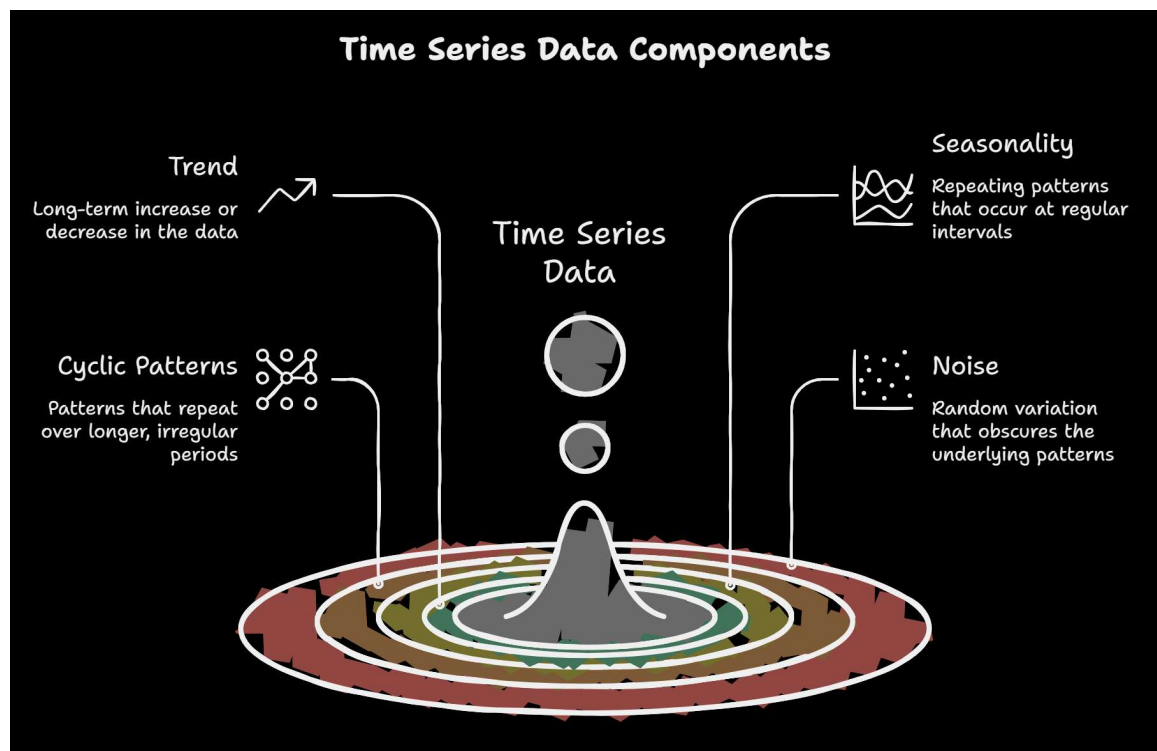


Patterns in Time Series

Time Series data contains patterns that help in forecasting.

Main Patterns:

- Trend (long-term movement)
- Seasonality (repeating pattern)
- Cyclic Pattern (irregular fluctuations)
- Noise (random variation)



Time Series Models

Time Series Models are used to learn patterns from past data and predict future values.

Types of Time Series Models

1. Statistical Models

- AR (Autoregressive – uses past values)
- MA (Moving Average – uses past errors)
- ARMA (combines AR + MA for stationary data)
- ARIMA (handles non-stationary data using differencing)
- ARIMAX (ARIMA with external variables)

- SARIMA (handles seasonality)
- SARIMAX (SARIMA with external/exogenous variables)

2. Machine Learning Models

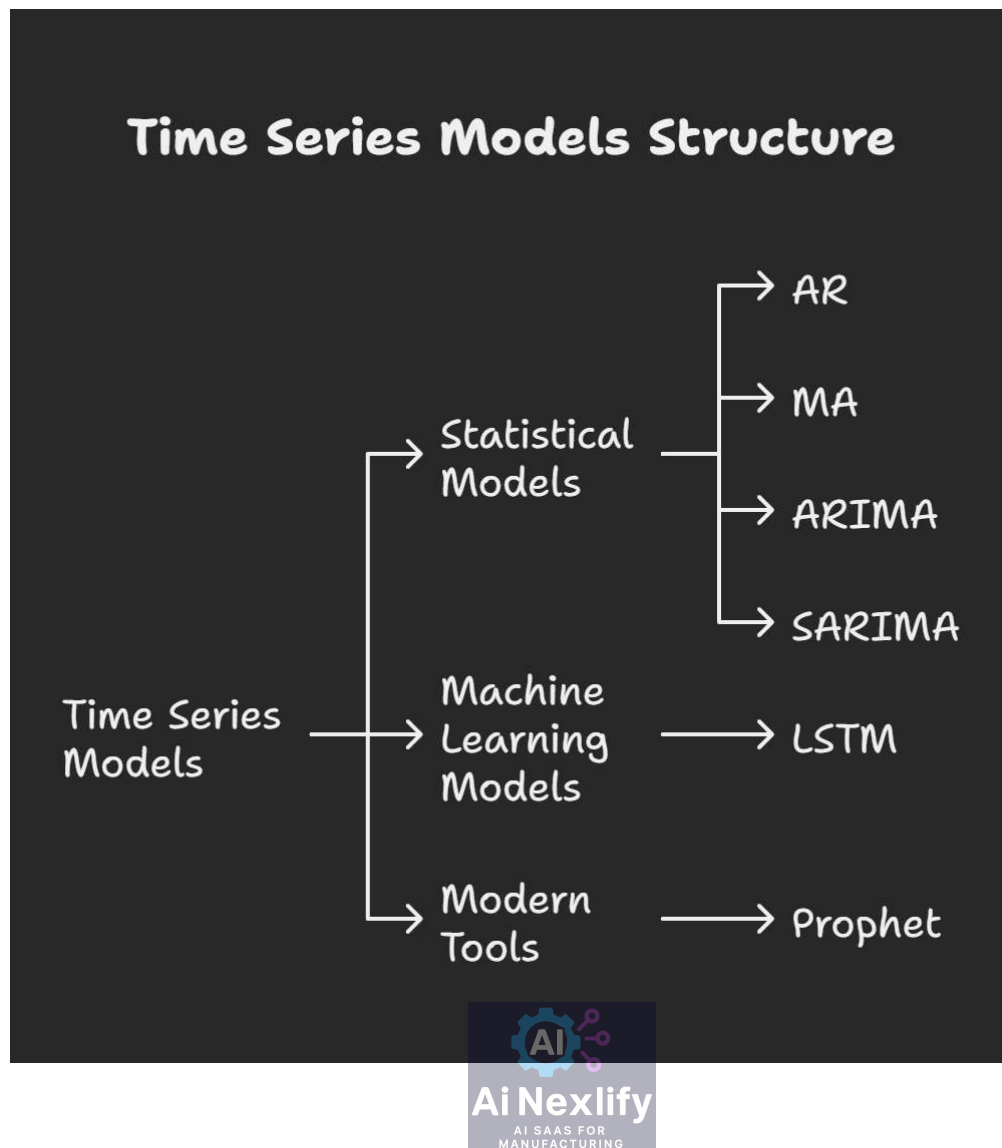
- XGBoost (tree-based boosting model using lag features)

3. Deep Learning Models

- RNN (Recurrent Neural Network – basic sequence model)
- LSTM (captures long-term dependencies)
- GRU (efficient version of LSTM)
- Autoencoder (learns compressed representations for anomaly detection)
- VAE (Variational Autoencoder – probabilistic representation learning)

4. Modern Tools / Frameworks

- Prophet (handles trend and seasonality automatically)



What is Seasonality?

Seasonality means a pattern that repeats at regular time intervals.

This pattern occurs because of external factors like:

- Weather
- Festivals
- Business cycles

Example:

- Ice cream sales increase every summer
- E-commerce sales increase during festivals (Diwali, Christmas)

Key Idea:

If something repeats after a fixed time period → it is seasonality

How to Identify Seasonality?

- Repeated peaks and drops at fixed intervals
- Same pattern every month / year

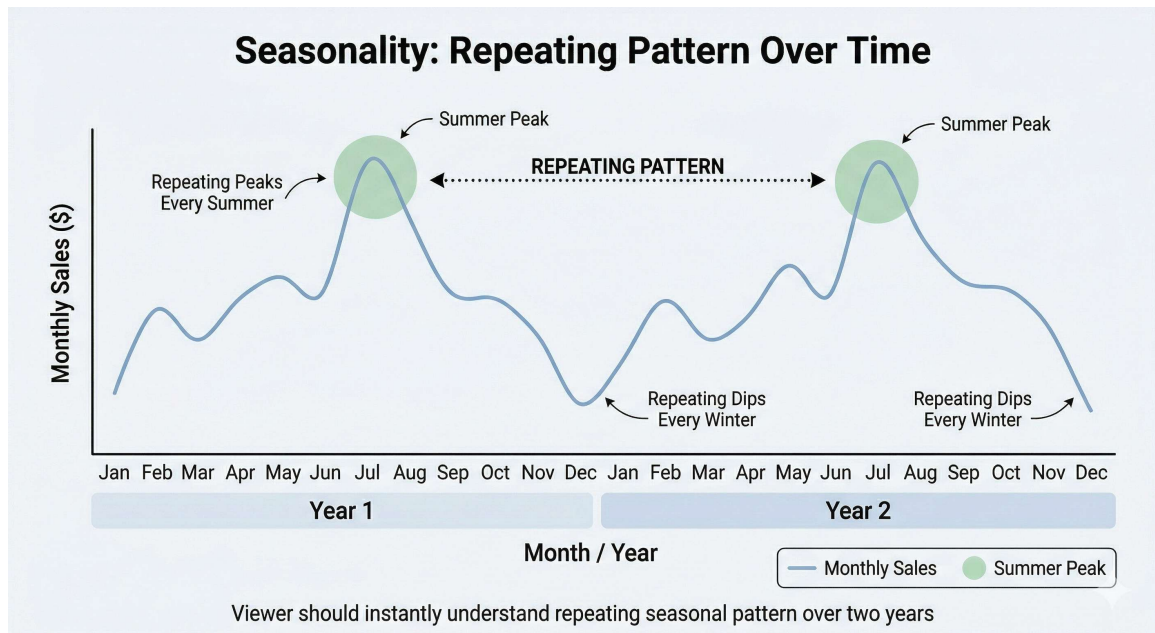
Real-Life Example:

Metric	Jan	Feb	Mar	Apr	May	Jun
Month	Jan	Feb	Mar	Apr	May	Jun
Sales	100	120	150	200	250	300

Metric	Jul	Aug	Sep	Oct	Nov	Dec
Month	Jul	Aug	Sep	Oct	Nov	Dec
Sales	320	300	250	200	180	150

Pattern repeats every year → Seasonality





What is Stationary Data?

A time series is called stationary if its statistical properties do not change over time.

These properties include:

- Mean (average)
- Variance (spread)
- Autocorrelation (relationship between values)

Simple Understanding:

Stationary data = stable behavior over time

Non-Stationary Data:

- Trend present (increasing or decreasing)
- Changing variance
- Seasonality present

Example:

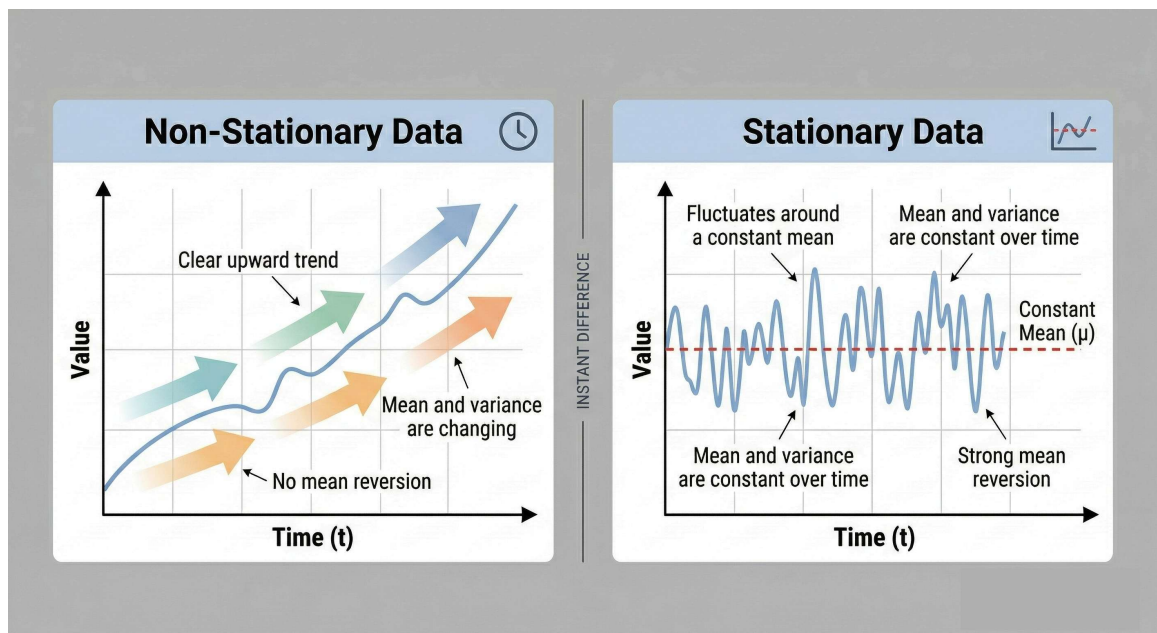
Non-Stationary: Sales: 100, 120, 150, 200, 260 (increasing trend)

Stationary: Values fluctuate around same level: 100, 105, 98, 102, 101

Why Stationarity is Important?

Most models like ARIMA require stationary data to work properly.





How to Check Stationarity?

1. Visual Method

Plot the data:

- If trend or seasonality is visible → non-stationary
- If stable → stationary

2. Statistical Test (ADF Test)

Augmented Dickey-Fuller Test

- Null Hypothesis: Data is non-stationary
- If p-value < 0.05 → Data is stationary

3. Rolling Statistics

Check:

- Rolling Mean
- Rolling Variance

If they are constant → stationary

ACF (Autocorrelation Function)

ACF measures correlation between: current value and past values

Example: How today's value is related to yesterday, day before yesterday, etc.

PACF (Partial Autocorrelation Function)

PACF measures: direct relationship between current value and lag, removing influence of intermediate lags

- Lag = How many steps back in time you're looking.

Simple Difference:

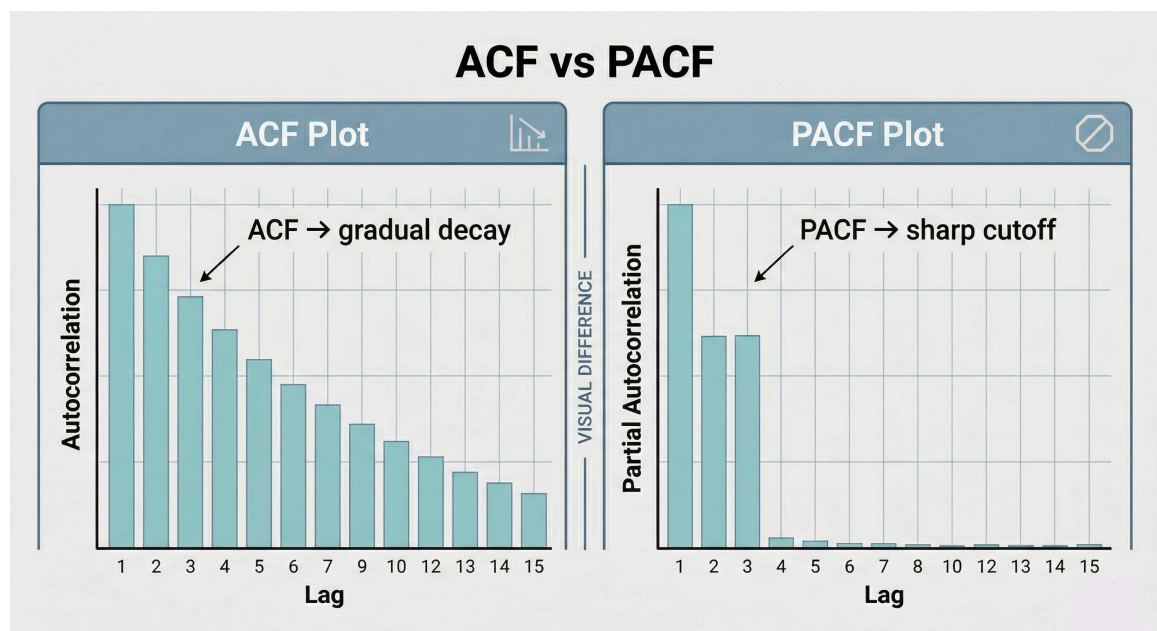
ACF → total correlation

PACF → direct correlation

Why Important?

Used to identify:

- AR order (from PACF)
- MA order (from ACF)



AR (Autoregressive Model)

AR model predicts future values using past values.

Formula Idea:

Current value depends on previous values



Example: $Y(t) = c + a_1Y(t-1) + a_2Y(t-2)$

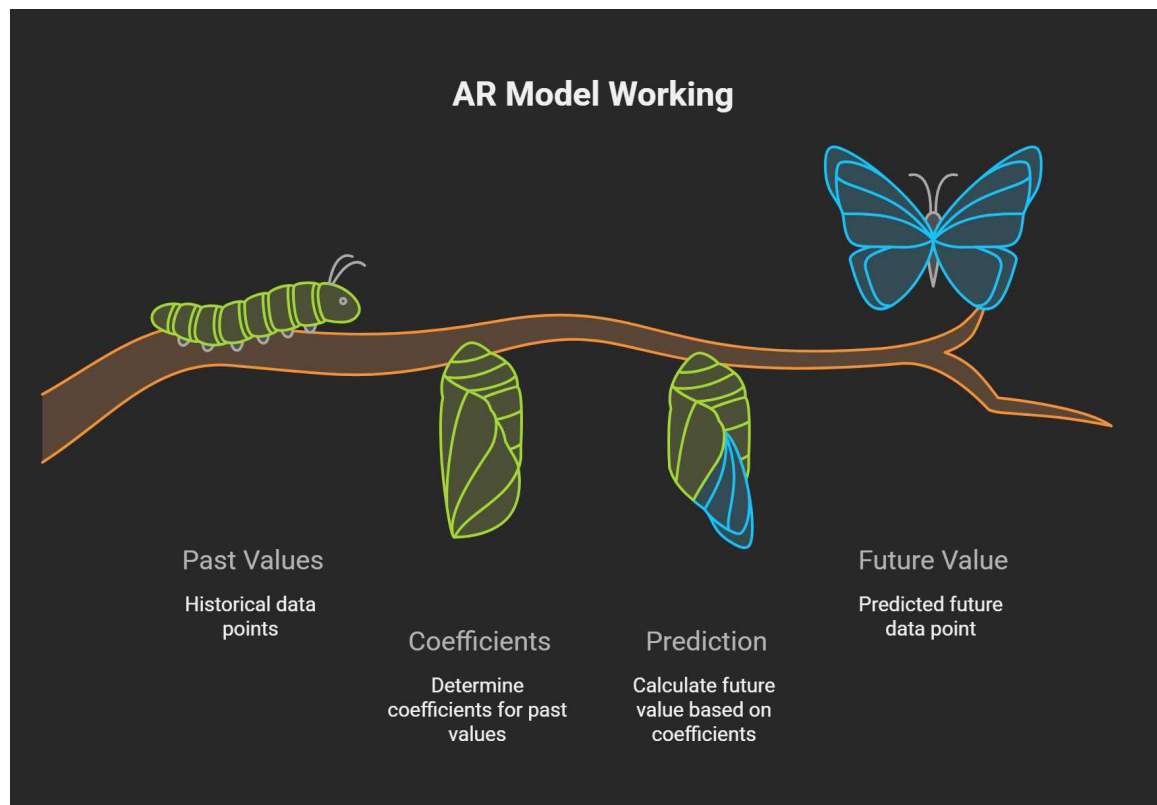
- $Y(t) \rightarrow$ "Today's Value" (The Goal)
- $c \rightarrow$ "Base Value" (The Starting Point)
- $a_1, a_2 \rightarrow$ "Weights" (The Importance)
- $Y(t-1) \rightarrow$ "Yesterday's Value" (Lag 1)
- $Y(t-2) \rightarrow$ "Day Before Yesterday's Value" (Lag 2)

Simple Example:

Today's temperature depends on yesterday's temperature

Key Point:

Uses past values only



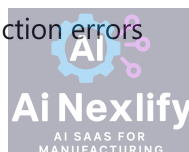
MA (Moving Average Model)

MA model uses past errors to make predictions.

Idea:

Current value depends on past prediction errors

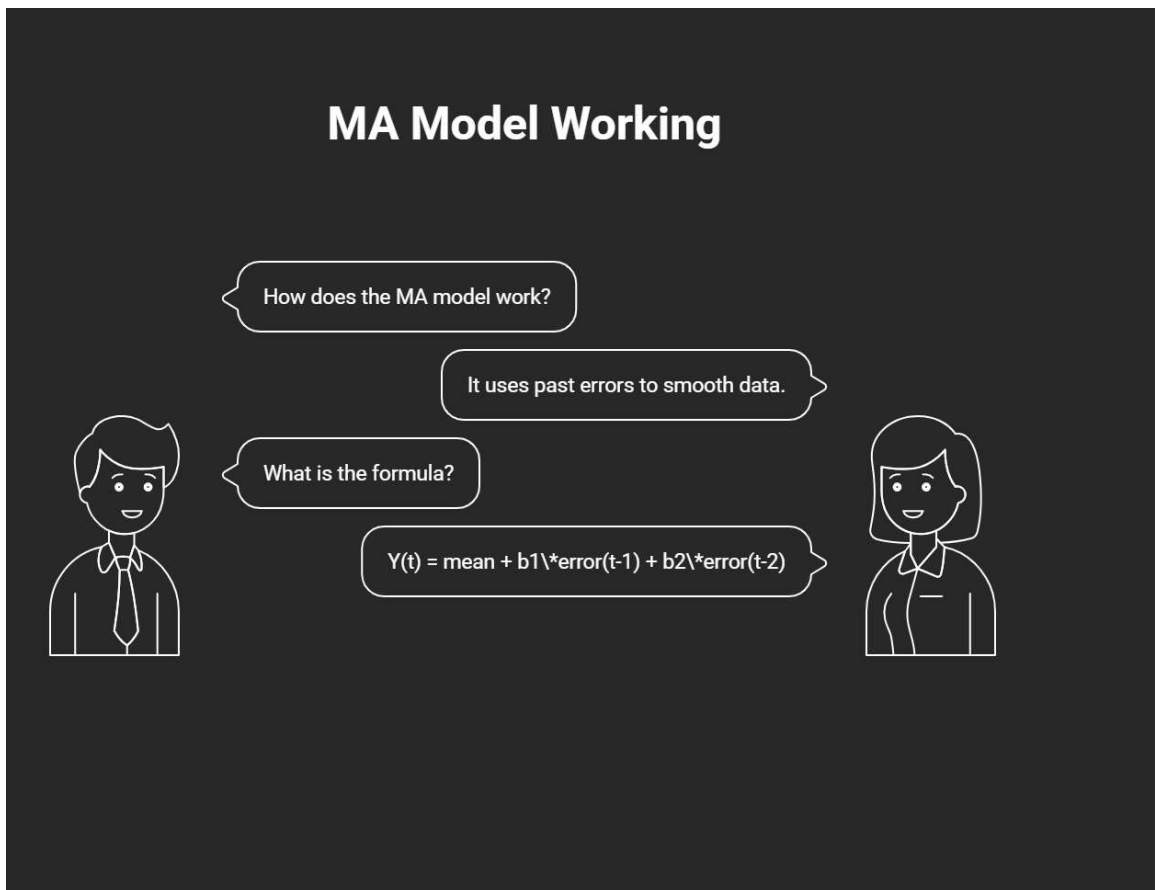
Example:



If yesterday's prediction had error, today's value adjusts using that error

Key Point:

Uses past errors, not past values



ARMA Model

Combination of:

- AR (past values)
- MA (past errors)

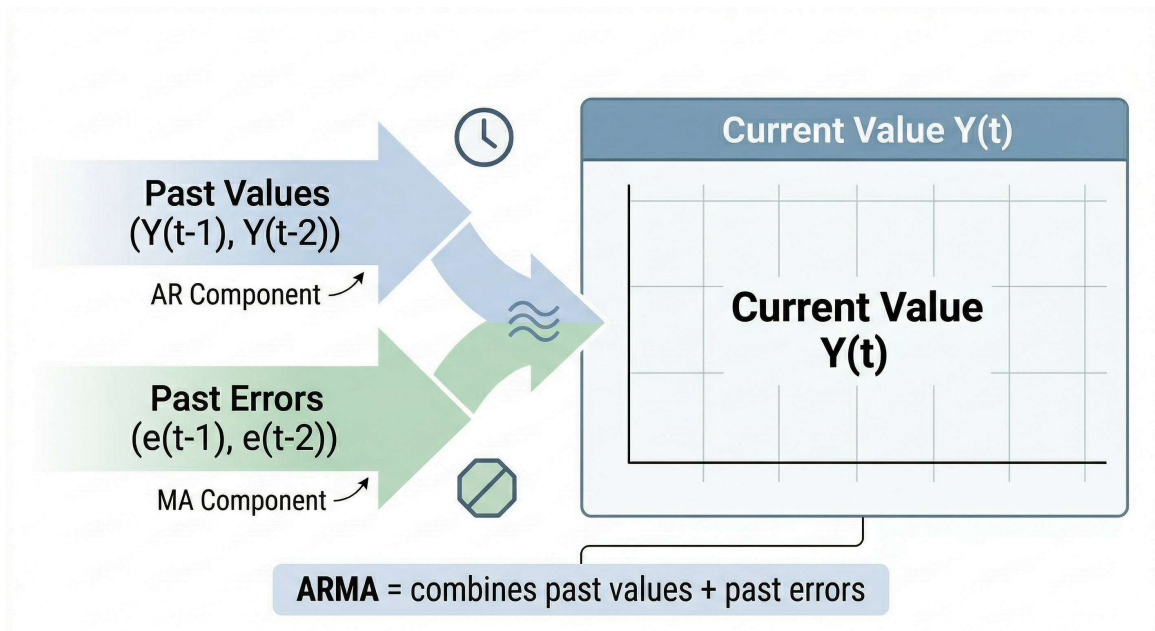
Use Case:

Works only for stationary data

Idea:

Uses both past values and past errors





ARIMA Model

ARIMA = AR + I + MA

Where:

- AR = past values
- I = differencing (to make data stationary)
- MA = past errors

Why ARIMA?

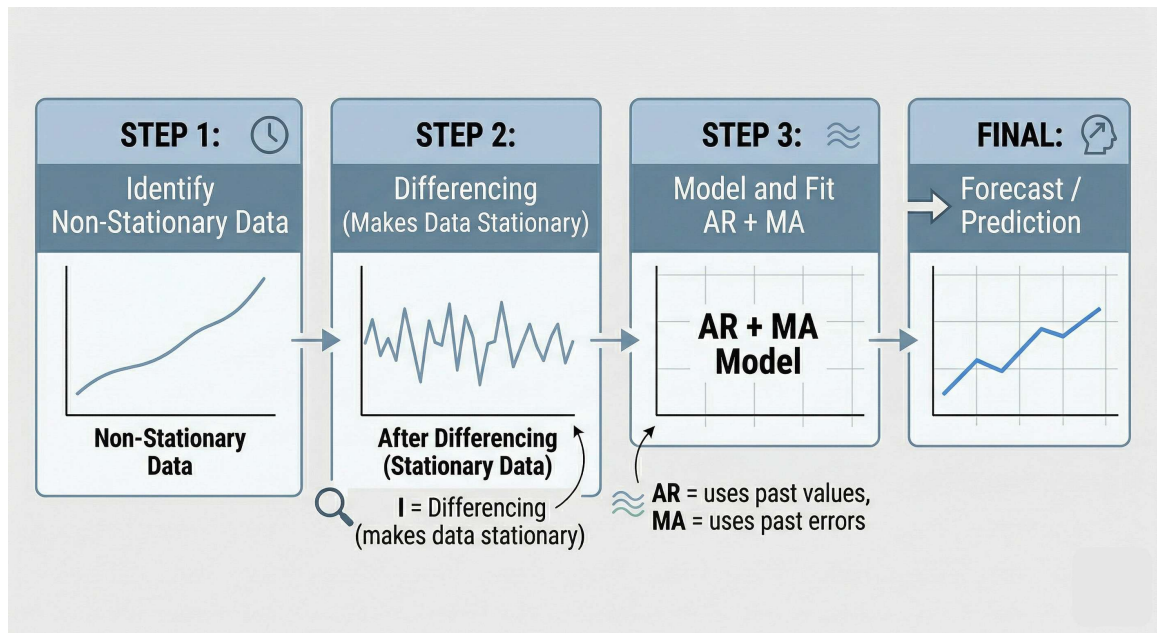
Real-world data is usually non-stationary

So we:

- Step 1: Make data stationary
- Step 2: Apply AR + MA

Example:

Sales data with trend → use ARIMA



ARIMAX Model

ARIMAX = ARIMA + External Variables

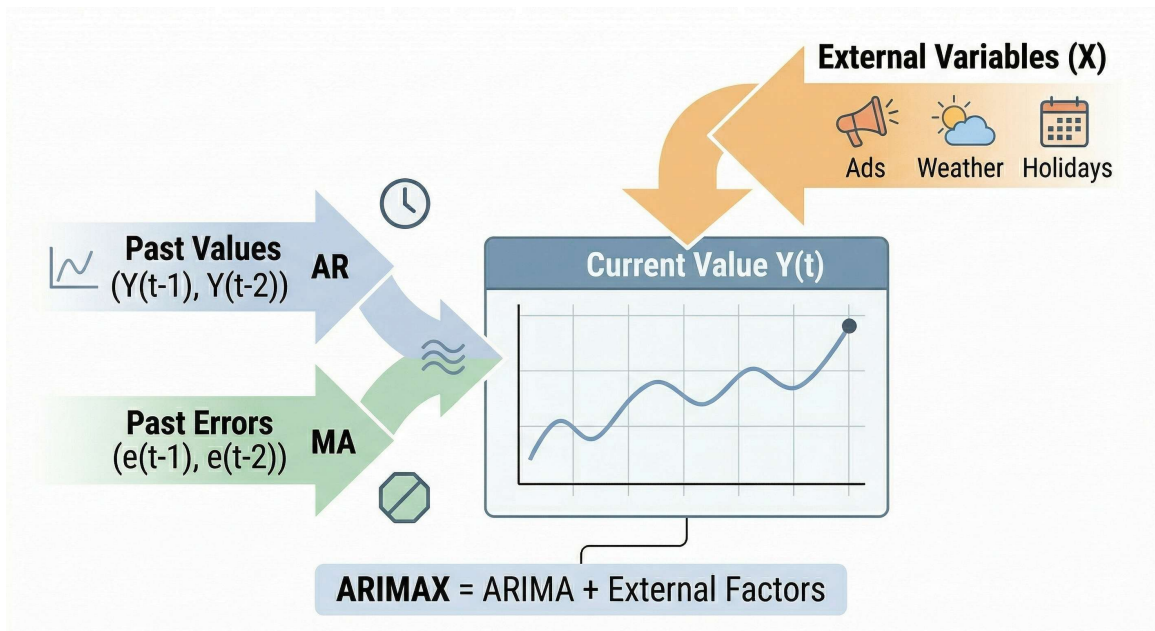
Example:

Predict sales using:

- Past sales
- Advertising spend

Key Idea:

Uses extra features along with time series



SARIMA Model

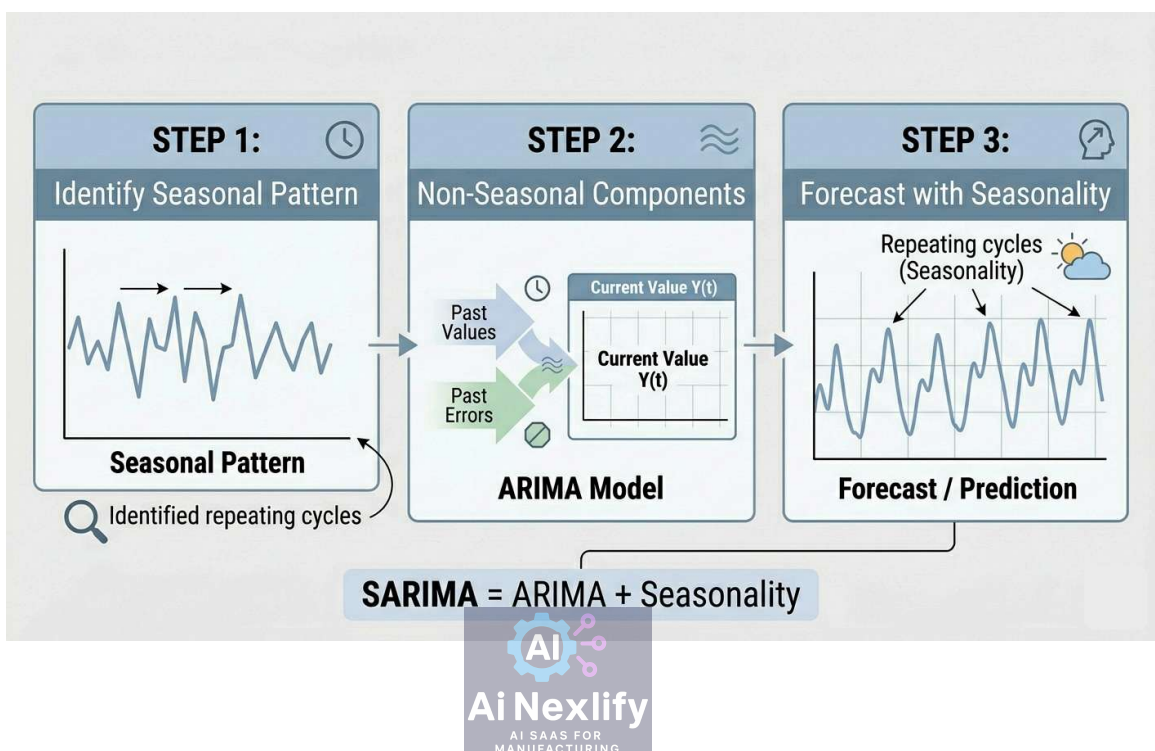
SARIMA = ARIMA + Seasonality

Example:

Monthly sales with yearly repeating pattern

Key Idea:

Handles repeating seasonal patterns



SARIMAX Model

SARIMAX = SARIMA + External Variables

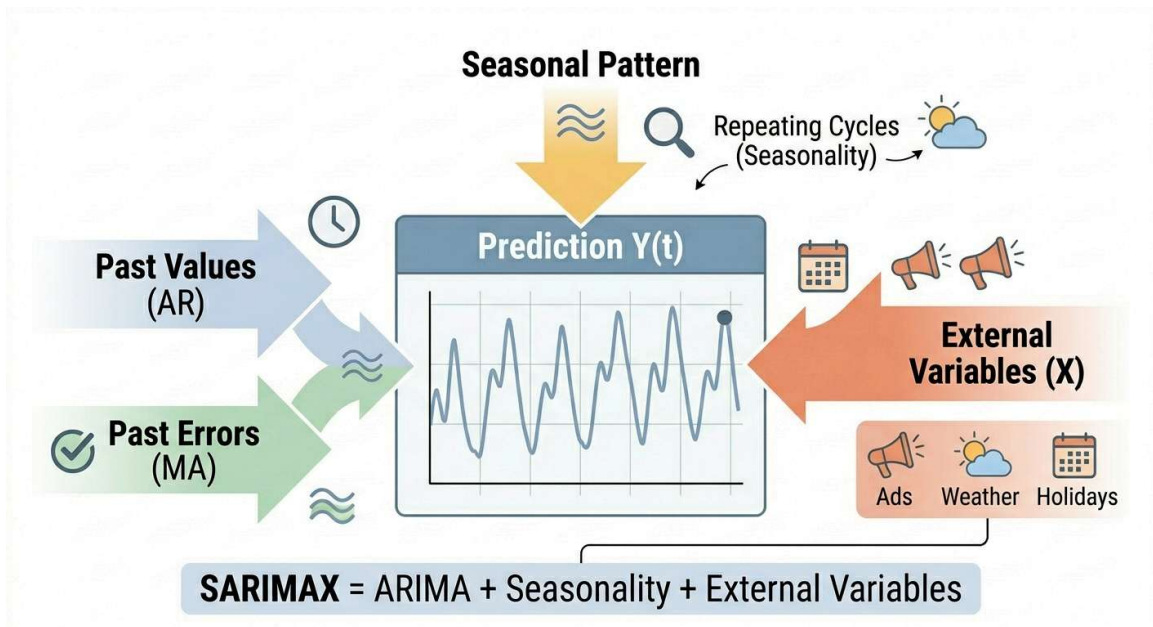
Example:

Predict seasonal sales using:

- Past sales
- Festival data
- Marketing campaigns

Key Idea:

Seasonality + external factors



Created by - Priyanka Deore & Kunal Mahadule